

Lesson 21: Taylor Polynomials

Taylor Polynomial Assessment Rubric

Criterion	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Mathematical Accuracy	All derivatives and coefficients are correct; the formula is applied flawlessly.	Minor calculation error in one derivative or coefficient, but the overall process is sound.	Multiple errors in derivatives, factorials, or powers of $(x - c)$.	Major conceptual errors; formula is incorrectly stated or applied.
Structure & Convergence	Correctly identifies the center c and the requested degree n .	Identifies the correct degree but may misplace the center in some terms.	Confuses Taylor and Maclaurin polynomials (center $c \neq 0$ vs $c = 0$).	Does not understand the relationship between degree and number of terms.
Error & Remainder Analysis	Accurately uses Taylor's Remainder Theorem to bound the maximum error.	Correctly sets up the remainder formula but has minor errors in the calculation of M .	Discusses error qualitatively but cannot calculate a numerical bound.	Does not attempt error analysis or does not understand "approximation".

Application & Context	Clearly explains why a higher degree improves accuracy.	Uses the polynomial to approximate a value but does not justify the degree chosen.	Can evaluate the polynomial for a given x but struggles with conceptual interpretation.	Cannot use the polynomial to approximate function values.
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